Mech HW

12 i) $Wd = \Delta GPE = 12 \times 5 \times 9.8 = 588\ J$

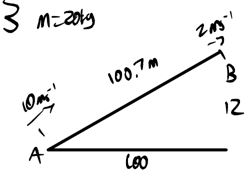
ii) $150 \times 6 = 900$ buckets
$= 4500$ kg of water

5m above ground level, $2ms^{-1}$ at top.

so 17m total. To move 4500 kg up 17m:

$Wd = GPE = 4500 \times 9.8 \times 17$
$= 749.7\ kJ$

13 $m = 20kg$



From A to B:

i) $KE_A = \frac{1}{2} \times 20 \times 10^2$
$= 1000\ J$

$KE_B = \frac{1}{2} \times 20 \times 2^2$
$= 40J$
$\therefore$ loss of $KE = 960J$

ii) Gain in GPE

$GPE = mg\Delta h$
$= 20 \times 9.8 \times 12$
$= 2352J$

iii) $Wd = F \times d$
$= 10 \times \sqrt{10144}$
$= 1007\ J$

iv) $W_d - 1000 = 2352 - 960$
$W_d = 2392$

$25\cos\alpha \times 100_m = 2392$

$\cos\alpha = \frac{2392}{100 \times 25}$
$\alpha = 16.9°$